Hat Finger

SKU:U074













Description

Hat Finger is a fingerprint recognition sensor. It integrates the FPC1020SC capacitive fingerprint recognition module, featuring multi-fingerprint entry, image processing, feature extraction, fingerprint matching, and search functions. It uses the UART communication protocol, has a compact design, and offers low power consumption, providing reliable security levels for projects while ensuring optimal user convenience. If you want to add biometric fingerprint recognition to your project with stable and reliable addition, verification, and management mechanisms, then **Hat Finger** is a great solution.

UART parameter settings:

- Baud rate (default: 19200bps)
- Start bit (1 bit)
- Stop bit (1 bit)
- Parity bit (none)

Features

- Fingerprint capacity: 150
- Uses capacitive matrix semiconductor fingerprint sensor
- Each pixel of the sensor has 256 grayscale levels of pixel quality
- Fingerprint data registration and comparison with minimal storage conditions: fingerprint template is 193 bytes
- 1:N recognition and 1:1 verification functions

- Supports 360-degree fingerprint rotation recognition
- Adjustable security level 0-9, default level 5
- Output format: username, image, feature value
- Feature value size: 193 bytes
- Communication interface: UART
- Communication baud rate: 9600-115200 (default is 19200)
- Operating temperature: -10° - 60°
- Relative humidity: 20% - 80%

Includes

- 1 x Hat Finger

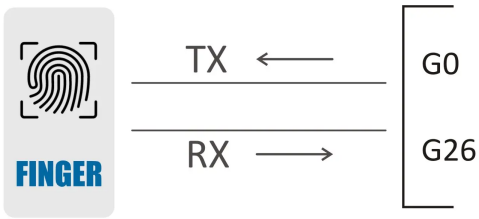
Applications

- Fingerprint attendance
- Password authentication replacement
- Identity recognition

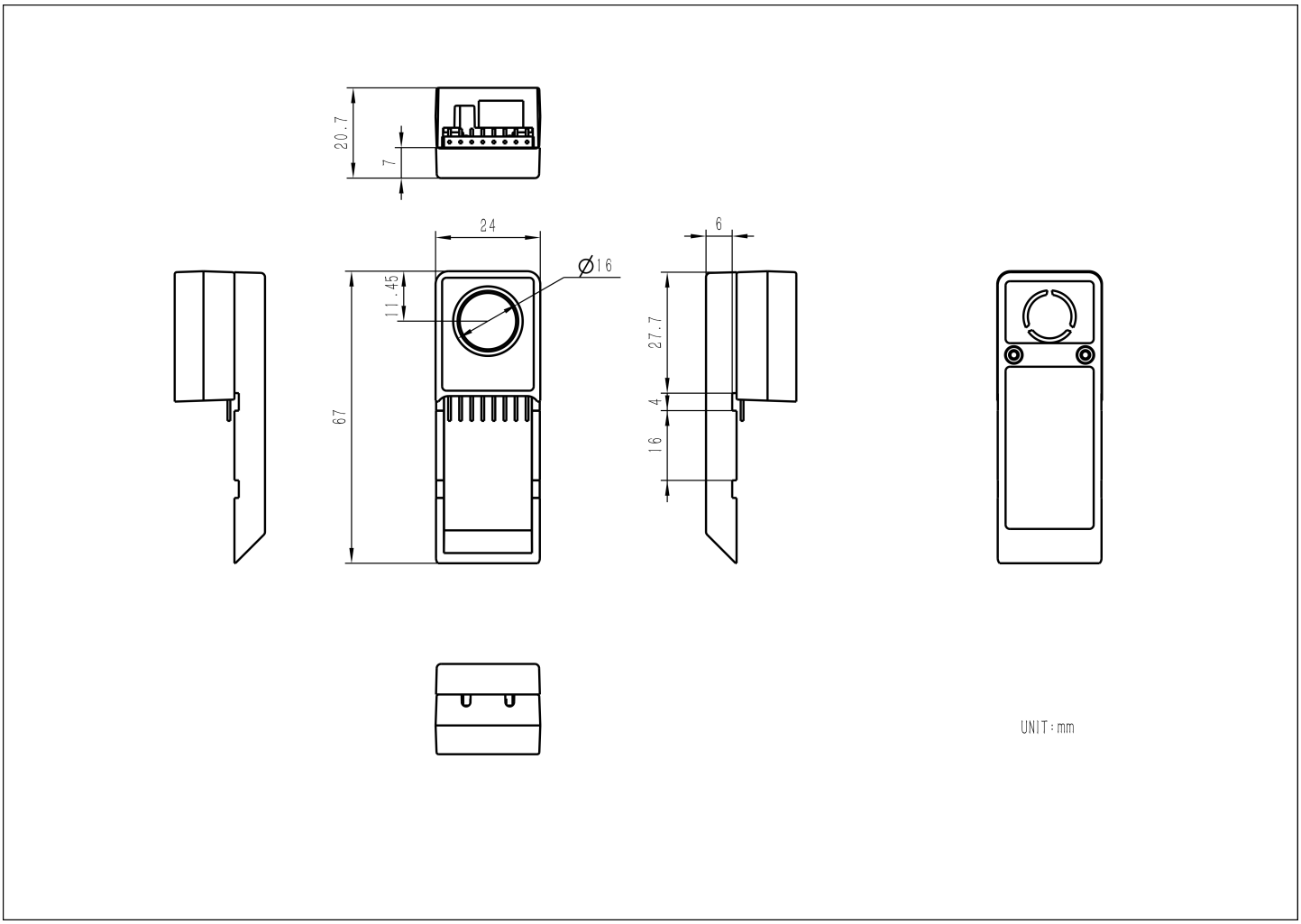
Specifications

Specification	Parameter
Net Weight	14g
Gross Weight	26g
Product Size	24 x 65 x 20mm
Package Size	75 x 46 x 29mm

Schematics



- Hat Finger Overall Design Block Diagram



Datasheets

FPC1020A

Softwares

Arduino

- Hat Finger - with M5StickC
- Hat Finger - with M5StickC-Plus

UiFlow1

- Hat Finger UiFlow1 Docs

UiFlow2

- Hat Finger UiFlow2 Docs

Protocol

- FINGER Serial Communication Protocol

Easyloader

Easyloader	Download Link	Notes
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Video

FINGER-HAT.mp4